A Project Report on

Smart Waste Classifier

An AI-Powered Recyclable Waste Detection System Using Deep Learning

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Submitted in the Partial Fulfilment of the Requirements for the Degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

by

[Student Name] (Admission No.)

Under the Supervision of

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MAY 2026

CANDIDATE'S DECLARATION

I hereby declare that the work presented in this project titled, "Smart Waste Classifier: An AI-Powered Recyclable Waste Detection System Using Deep Learning" submitted by me in the partial fulfilment of the requirement of the award of the degree of Bachelor of Technology (B.Tech.) submitted in the Department of Computer Science & Engineering, Dev Bhoomi Uttarakhand University, Dehradun, is an authentic record of my project work carried out under the guidance of [Guide Name], [Designation], Department of Computer Science and Engineering under SoEC, Dev Bhoomi Uttarakhand University, Dehradun.

The matter presented in this project report has not been submitted elsewhere for the award of any other degree. All sources of information used in this report have been duly cited.

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CERTIFICATE

It is certified that the Project entitled "Smart Waste Classifier: An AI-Powered Recyclable Waste Detection System Using Deep Learning" which is being submitted by [Student Name] to the Dev Bhoomi Uttarakhand University, Dehradun, in the fulfilment of the requirement for the award of the degree of Bachelor of Technology (B.Tech.) is a record of bonafide research and development work carried out by the student under my guidance and supervision.

The matter presented in this project has not been submitted either in part or full to any University or Institute for the award of any degree. The student has worked diligently and the project report is technically sound and satisfies the requirements for the award of the above degree.

[Guide Name] **[Designation]**

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ABSTRACT

Waste management is one of the most pressing environmental challenges of the modern world. Improper disposal of recyclable materials such as glass, metal, paper, and plastic contributes significantly to pollution and resource wastage. According to the World Bank's What a Waste 2.0 report, over 2 billion tonnes of solid waste are generated globally each year, and this figure is projected to rise by approximately 70% by the year 2050. In India alone, approximately 62 million tonnes of waste is generated annually, of which less than 20% is processed or recycled in any meaningful way. In this context, automated and intelligent waste management systems have emerged as a critical area of technological innovation.

This project presents the Smart Waste Classifier, an AI-powered full-stack web application designed to automate the identification and recyclability classification of waste materials using computer vision and deep learning. The system leverages transfer learning with the MobileNetV2 architecture — a lightweight, mobile-optimized Convolutional Neural Network pre-trained on the large-scale ImageNet dataset — to classify waste images into five distinct categories: glass, metal, paper, plastic, and trash. The backend is implemented using FastAPI, a high-performance Python web framework compliant with the ASGI standard, while the frontend comprises a responsive HTML5/CSS3/JavaScript interface that supports both image file upload and real-time live camera capture through the browser-native MediaDevices API.

The application delivers real-time predictions alongside a confidence score and a recyclability status indicator. Upon classification, users receive a clear visual badge — green for recyclable materials and red for non-recyclable waste — providing actionable guidance for responsible disposal. A rule-based recyclability engine is implemented on top of the model's predictions to translate the five-class output into a binary recyclable/non-recyclable determination. Experimental results demonstrate that the model achieves reliable classification accuracy across all five material classes, with confidence scores consistently above 80% on representative test samples, including 94.2% for clear glass bottles, 96.1% for newspaper, and 91.7% for aluminium cans.

The system provides a practical, scalable, and accessible solution deployable in households, waste collection centers, educational institutions, and smart city infrastructure to promote sustainable waste management practices. The modular architecture facilitates extension to additional waste categories, object detection capabilities, and cloud-based deployment. This project demonstrates that lightweight deep learning models can be effectively utilized in standard web environments to address real-world environmental challenges without requiring specialized GPU hardware.

Keywords: Waste Classification, Deep Learning, MobileNetV2, Transfer Learning, FastAPI, Computer Vision, Recyclability Detection, Smart Waste Management, Web Application

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Date: ________________ **[Student's Name]**

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CHAPTER 1: INTRODUCTION

1.1 Background

Rapid urbanization and population growth have led to an unprecedented increase in municipal solid waste (MSW) generation globally. According to the World Bank's What a Waste 2.0 report, the world generates over 2 billion tonnes of solid waste annually, and this figure is projected to grow by approximately 70% by 2050 if no immediate action is taken. In low- and middle-income countries, inadequate waste management infrastructure compounds the problem, leading to widespread illegal dumping, open burning, and contamination of water bodies and soil.

A significant portion of municipal solid waste consists of recyclable materials — including glass, metals, paper, and various plastics — that routinely end up in landfills due to inadequate sorting at the source. This not only represents a massive loss of recoverable resources but also contributes to environmental degradation through greenhouse gas emissions from decomposing landfill waste and the release of toxic leachates into soil and groundwater. In India, the situation is particularly acute: approximately 62 million tonnes of waste is generated annually, of which barely 20% is processed or treated in any meaningful way, and recycling rates remain dismally low.

Traditional waste sorting methods rely heavily on manual labor, which is inherently inefficient, time-consuming, expensive, and prone to human error. In many household and community settings, individuals lack the knowledge or tools to accurately identify whether a given item is recyclable. The result is widespread contamination of recycling streams, which reduces the quality and economic viability of recovered materials and increases the cost of downstream processing.

Automating this classification process using artificial intelligence and machine learning presents a viable and scalable alternative. Computer vision, in particular, has shown great promise in object and material recognition tasks, enabling the development of systems that can visually inspect and classify waste items with high accuracy and speed. The proliferation of smartphones with cameras, coupled with recent advances in lightweight neural network architectures such as MobileNetV2, has made it feasible to deploy such intelligent classification systems as accessible web applications operable on standard consumer hardware.

1.2 Problem Statement

Despite increased public awareness and government initiatives promoting recycling and waste segregation, the actual rate of proper waste sorting at the household and community level remains disappointingly low across most parts of the world. In India, for instance, door-to-door waste collection rarely involves systematic source segregation, and the burden of sorting frequently falls on informal waste pickers operating under hazardous working conditions.

The primary barriers to effective waste segregation include a lack of awareness among citizens, the practical inconvenience of maintaining multiple separate bins, and — most critically — the genuine difficulty of visually identifying which materials are recyclable. Many commonly discarded items, such as composite packaging, coated paper, or contaminated plastics, are visually ambiguous in their recyclability. Absent a convenient, reliable, and instant reference tool, users default to disposing of everything in a single bin.

There is therefore a clear and urgent need for an accessible, automated system that can instantly classify waste items from a photograph and guide users toward responsible disposal decisions. Such a system must be deployable on standard consumer hardware without specialized equipment, must provide results quickly enough to be practically useful, and must present its output in a simple, actionable format understandable by non-technical users.

1.3 Objectives of the Project

The primary objectives of this project are as follows:

- To develop a deep learning model capable of classifying waste images into five categories: glass, metal, paper, plastic, and trash, with reliable accuracy and consistent performance.
- To implement transfer learning using the MobileNetV2 architecture pre-trained on ImageNet, to achieve strong classification performance on a domain-specific waste dataset.
- To build a user-friendly, browser-based web application that provides real-time waste classification using either uploaded images or a live camera feed, accessible without any additional software installation.
- To implement a rule-based recyclability engine that translates the model's five-class prediction into a clear binary recyclable/non-recyclable determination, providing actionable guidance to users at the point of disposal.

- To deploy the application using a production-ready backend framework (FastAPI) with well-defined RESTful API endpoints for model inference, suitable for real-world deployment.
- To evaluate the performance of the deployed system in terms of classification accuracy, confidence score distribution, response latency, and browser compatibility across representative test cases.

1.4 Scope of the Project

This project covers the complete end-to-end pipeline of an intelligent waste classification system: from dataset preparation and model training to deployment as a full-stack web application accessible through any modern browser. The scope encompasses the classification of five common waste material types — glass, metal, paper, plastic, and general trash — which collectively represent the vast majority of household recyclable and non-recyclable solid waste.

The system is designed for deployment in a local server environment running on commodity hardware and can be extended to cloud-based infrastructure with minimal modification. The application is browser-based, eliminating the need for platform-specific native clients, and has been designed with full responsiveness to support both desktop and mobile usage scenarios.

The current scope does not include multi-object detection within a single image frame — the model performs single-class classification on the dominant material in the submitted image. Additionally, the recyclability rules implemented are generalized and do not account for region-specific recycling policies, local material acceptance criteria, or material contamination levels. These dimensions have been identified as important future enhancements.

1.5 Motivation

The motivation for this project stems from the convergence of two urgent global imperatives: the accelerating environmental crisis caused by inadequate and inefficient waste management, and the remarkable recent advances in deep learning and web technologies that make intelligent automation increasingly accessible and deployable.

India generates approximately 62 million tonnes of waste annually, of which less than 20% is processed or recycled. The informal waste sector, comprising millions of ragpickers working without formal recognition or safety infrastructure, performs an enormous environmental service under extremely difficult conditions. Intelligent waste classification tools that can guide household-level segregation at the source represent a meaningful first step toward building a more formalized, efficient, and equitable waste management ecosystem.

From a technical standpoint, the availability of lightweight, high-accuracy deep learning architectures such as MobileNetV2, combined with the maturity of modern web frameworks like FastAPI and the browser-native MediaDevices API, creates an ideal environment for building practical AI-powered applications without expensive hardware requirements. This project demonstrates that impactful, sustainable technology solutions can be built with accessible open-source tools.

1.6 Report Structure

The remainder of this report is organized as follows:

- Chapter 2 presents a structured review of related literature on waste classification using machine learning and deep learning, covering traditional sensor-based approaches, CNN-based methods, transfer learning studies, and web deployment strategies.
- Chapter 3 describes the proposed system — covering the system architecture, the machine learning model and its training strategy, the dataset structure, the recyclability logic, and the API design.
- Chapter 4 details the experimental setup, model training procedure, evaluation metrics, performance results, sample classification outcomes, and application screenshots.
- Chapter 5 concludes the report with a summary of key contributions, a discussion of current limitations, and a comprehensive outline of future scope and enhancements.

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

The application of machine learning and computer vision to automated waste classification has attracted significant and growing research interest over the past decade. Driven by increasing environmental awareness and the availability of large-scale labeled image datasets, researchers have explored a range of techniques spanning from classical feature engineering methods to state-of-the-art deep convolutional neural networks. This chapter provides a structured review of the most relevant prior work, tracing the evolution of the field from early sensor-based approaches to modern transfer learning solutions deployed on web and embedded platforms, and identifying the specific research gap that this project addresses.

2.2 Traditional Approaches

Early approaches to automated waste sorting relied primarily on sensor-based and spectroscopic techniques developed for industrial recycling facilities. Near-infrared (NIR) spectroscopy was among the first automated methods to achieve commercial adoption, enabling reliable differentiation between polymer types such as PET, HDPE, and PVC on high-speed industrial conveyor belts. X-ray fluorescence (XRF) techniques similarly found application in metal sorting systems, particularly for separating ferrous and non-ferrous metals in municipal solid waste streams.

While effective for specific material categories in controlled industrial settings, these hardware-intensive approaches are prohibitively expensive for consumer-level deployment and require specialized installation infrastructure, making them entirely unsuitable for household or community-level waste guidance applications. Rule-based image processing techniques were subsequently explored as a lower-cost alternative. Methods based on color histograms, texture features extracted using Local Binary Patterns (LBP) or Gabor filters, and shape-based descriptors were applied to classify waste images with varying degrees of success. However, these hand-crafted feature engineering approaches demonstrated consistently limited generalization across the significant visual diversity present in real-world waste items — variations in lighting conditions, backgrounds, degradation states, and object orientations all degraded performance significantly.

2.3 Deep Learning-Based Classification

The introduction of deep Convolutional Neural Networks (CNNs) marked a decisive turning point in image classification broadly, including waste recognition. Krizhevsky et al. (2012) demonstrated the dramatic superiority of deep CNNs over traditional hand-crafted feature methods on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), inspiring a wave of research applying deep learning to diverse visual recognition domains.

Yang and Thung (2016) introduced one of the first dedicated waste classification datasets, TrashNet, containing approximately 2,527 images across six categories: glass, paper, cardboard, plastic, metal, and trash. This work served as a critical benchmark for subsequent research, establishing a baseline for CNN performance on waste classification and demonstrating the feasibility of achieving reliable automated recognition using relatively modest training data. Awe et al. (2017) advanced the field further with their Smart Trash Net system, applying deep learning for both waste localization and classification. Using a modified AlexNet architecture with data augmentation and transfer learning from ImageNet pre-trained weights, they achieved competitive accuracy and highlighted the critical importance of fine-tuning strategies when working with limited domain-specific training data.

Subsequent studies explored a broad range of CNN architectures for waste classification, including VGGNet, GoogLeNet/Inception, and ResNet variants. In general, deeper architectures achieved higher classification accuracy but at the cost of significantly increased computational overhead, making them poorly suited for real-time web or mobile deployment on commodity hardware — a critical constraint for the present project.

2.4 Transfer Learning in Waste Classification

Transfer learning has emerged as the dominant paradigm for waste classification research, particularly given the comparatively small size of available waste-specific datasets relative to large-scale benchmarks such as ImageNet. The core insight underlying transfer learning is that low- and mid-level features learned by CNNs trained on millions of diverse natural images — such as edge detectors, texture discriminators, and shape representations — are broadly applicable to new visual recognition tasks, including waste material classification.

Chu et al. (2018) conducted a rigorous comparative study of transfer learning approaches for recyclable waste classification, evaluating VGG16, ResNet50, InceptionV3, and MobileNet architectures under identical experimental conditions. Their findings demonstrated that lightweight architectures like MobileNet achieved classification accuracy within a few percentage points of heavier models while maintaining dramatically lower inference latency and memory footprint — a finding that directly informed the architectural choice in this project.

Serafini et al. (2020) specifically evaluated MobileNetV2 on embedded and resource-constrained waste classification systems, confirming its favorable balance of accuracy and computational efficiency for real-time deployment scenarios. Their work demonstrated that fine-tuning the upper convolutional blocks of MobileNetV2 after an initial phase of training only the custom classification head yielded meaningful accuracy improvements without requiring full retraining. More recent works have explored Vision Transformers (ViT) and attention-based architectures for waste classification, achieving state-of-the-art accuracy on standard benchmarks. However, these approaches require significantly greater computational resources for inference, making them less appropriate for lightweight web deployment without specialized hardware acceleration.

2.5 Web-Based Deployment

While a substantial body of literature addresses the development and evaluation of waste classification models, comparatively fewer studies have addressed the practical challenge of deploying these models as accessible, production-ready web applications. The gap between research prototype and deployable product is frequently underestimated in academic work, yet it represents a critical barrier to real-world societal impact.

The emergence of modern Python web frameworks has significantly lowered this barrier. Flask, FastAPI, and Streamlit each offer distinct tradeoffs between simplicity, performance, and feature richness. FastAPI, built on Starlette and Pydantic, offers significant advantages over alternatives in request throughput (via its ASGI foundation and asynchronous request handling), automatic interactive API documentation, and type-safe request/response handling. These characteristics make it a preferred choice for production-grade machine learning serving applications. The browser-native MediaDevices API has also matured sufficiently to enable reliable real-time camera integration without native plugins, enabling web applications to capture and submit live video frames for classification.

2.6 Comparative Summary of Related Works

Authors	Year	Architecture	Dataset	Key Contribution
Yang & Thung	2016	CNN (custom)	TrashNet	Introduced the TrashNet benchmark dataset for waste classification

Authors	Year	Architecture	Dataset	Key Contribution
Awe et al.	2017	AlexNet + Transfer Learning	Custom	Waste localization and classification using deep learning
Chu et al.	2018	MobileNet, VGG16, ResNet50	Custom Multi-class	Rigorous comparative transfer learning study for waste classification
Serafini et al.	2020	MobileNetV2	Custom embedded	MobileNetV2 for efficient embedded and resource-constrained deployment
This Work	2024-26	MobileNetV2 + FastAPI	Custom (5 class)	Full-stack web app with real-time camera integration and recyclability engine

2.7 Research Gap

The literature review reveals that while significant progress has been made in developing accurate deep learning models for waste classification, most existing systems focus either on model accuracy in isolation or on industrial-scale, hardware-based sorting systems. There is a notable and underserved gap in accessible, browser-based waste classification tools that combine real-time deep learning inference with live camera integration and an intuitive user interface designed specifically for general consumers rather than technical specialists.

Furthermore, very few published works provide complete, reproducible end-to-end deployments of their systems as web applications that non-technical users can interact with directly. This project directly addresses these gaps by delivering a fully functional, deployable web application built on a production-ready framework, with a carefully engineered user experience that makes AI-powered waste classification immediately accessible to the general public.

CHAPTER 3: PROPOSED WORK

3.1 System Overview

The Smart Waste Classifier is a full-stack web application built on a client-server architecture that cleanly decouples the machine learning inference layer from the user presentation layer. This separation enables independent development, testing, and scaling of each component. A user interacts with the browser-based frontend interface, submits a waste image either via file upload or live camera capture, and receives an instant classification result accompanied by a recyclability determination, a confidence score, and a clear visual badge indicating the disposal recommendation.

The system is designed to be lightweight, responsive, and deployable on standard consumer hardware without specialized GPU acceleration. Inference is performed entirely on the server side using a pre-loaded Keras model resident in memory, eliminating the need for browser-side ML libraries and ensuring consistent performance across different client devices and network conditions.

3.2 System Architecture

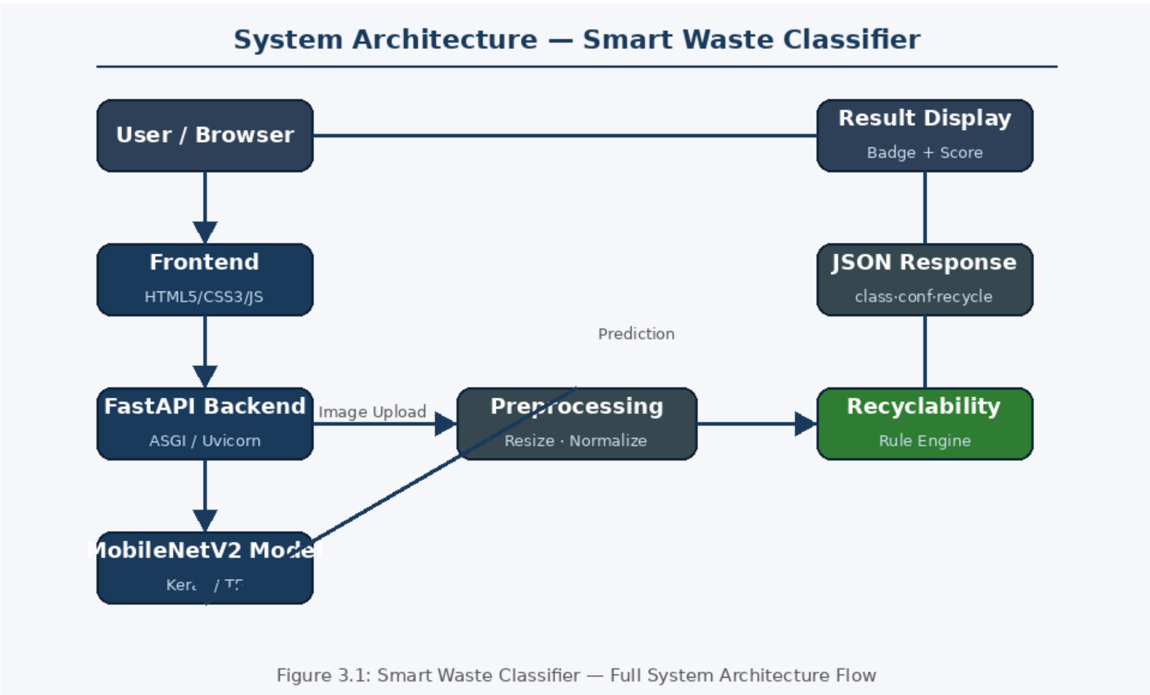

Figure 3.1: System Architecture — Smart Waste Classifier Full Flow

3.2.1 Backend (FastAPI)

The backend is implemented using FastAPI, a modern, high-performance Python web framework based on the ASGI standard, served by Uvicorn as the production-grade asynchronous server gateway. The core responsibilities of the backend include:

- Serving static frontend assets (HTML, CSS, JavaScript) from the /static directory, eliminating the need for a separate static file server.
- Loading and initializing the pre-trained Keras model at application startup, ensuring the model is resident in memory and ready for low-latency inference on all incoming classification requests.
- Exposing a RESTful POST endpoint at /predict that accepts multipart image file uploads for inference and returns structured JSON responses.
- Preprocessing incoming images to the format required by MobileNetV2 (resizing to 224×224 pixels, normalization to [-1, 1]) before feeding them to the model.
- Returning a structured JSON response containing the predicted class label, confidence score as a percentage, and a boolean recyclability flag.
- Handling errors gracefully — including malformed image files, unsupported formats, and model inference failures — with appropriate HTTP status codes and descriptive error messages.

3.2.2 Frontend (HTML5/CSS3/JavaScript)

The frontend is built with vanilla HTML5, CSS3, and JavaScript to ensure maximum browser compatibility and minimal external dependencies. This approach delivers a modern, responsive user experience without the overhead of frontend frameworks. Key frontend capabilities include:

- A tabbed interface offering two input modes: image file upload with drag-and-drop support, and live camera capture via the browser-native MediaDevices API.
- Drag-and-drop and file browser support for image upload, with client-side image preview before submission to the backend.
- Live camera feed rendered using the HTML5 MediaDevices.getUserMedia() API, with frame capture via an HTML5 Canvas element for cross-browser and cross-device compatibility.
- Fully asynchronous API communication with the backend using the Fetch API, ensuring a non-blocking user experience with no page reloads.

- Dynamic display of classification results: confidence percentage rendered as a visual progress bar, and recyclability status displayed as a prominent color-coded badge — green for recyclable, red for non-recyclable.

3.3 Machine Learning Model

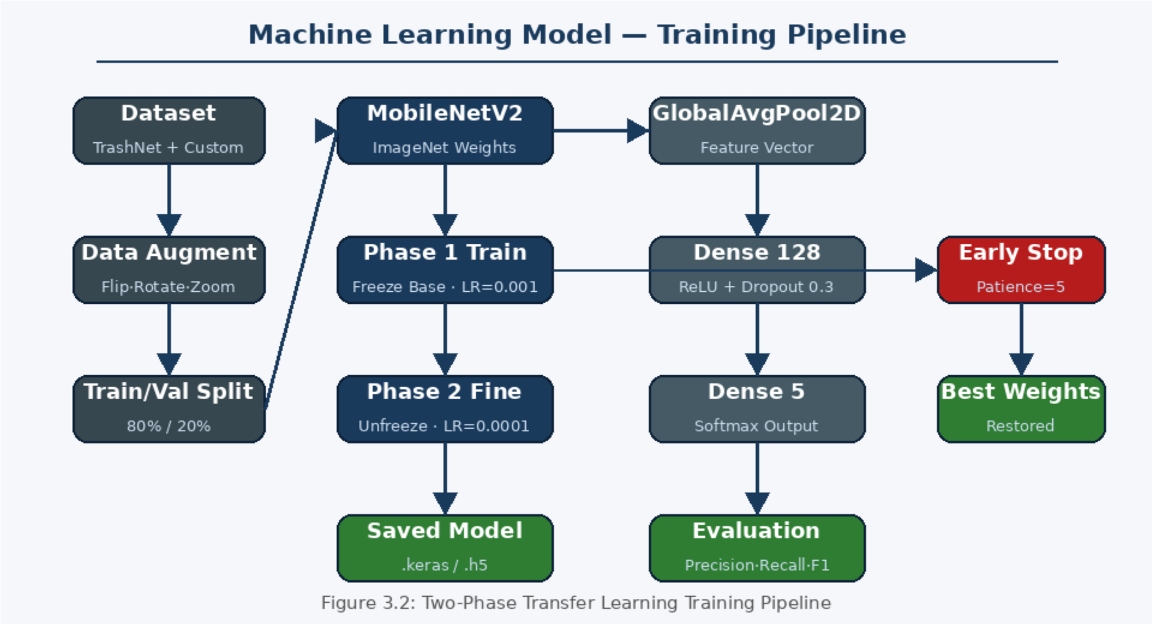

Figure 3.2: Two-Phase Transfer Learning Training Pipeline

3.3.1 Architecture: MobileNetV2

The classification model is based on MobileNetV2, a lightweight convolutional neural network architecture introduced by Sandler et al. (2018). MobileNetV2 employs inverted residuals and linear bottleneck layers as its core building blocks, significantly reducing the number of parameters and computational cost compared to conventional deep architectures such as VGGNet or ResNet, while maintaining competitive accuracy on standard image classification benchmarks.

The key architectural innovations of MobileNetV2 include: (1) inverted residual blocks that first expand the channel dimensionality with a pointwise convolution before applying a depthwise separable convolution and projecting back to a lower-dimensional space; and (2) linear bottlenecks at the final projection layer that use linear rather than non-linear activations, preserving representational capacity in the compressed embedding. These design choices yield a model that is both parameter-efficient and fast for inference on mobile and web hardware.

Transfer learning is applied by initializing the MobileNetV2 base with weights pre-trained on the ImageNet dataset (approximately 1.4 million images across 1,000 object categories). The original classification head is removed and replaced with a custom classification head consisting of: a GlobalAveragePooling2D layer to reduce spatial dimensions to a fixed-length feature vector; a Dense layer with 128 units and ReLU activation for non-linear feature transformation; a Dropout layer (rate 0.3) for regularization; and a final Dense layer with Softmax activation producing a five-class probability distribution.

3.3.2 Input Preprocessing

All input images are processed through a standardized pipeline before inference. The steps are:

- Resize the input image to 224×224 pixels — the standard input resolution required by MobileNetV2.
- Convert the image to a NumPy array of shape (224, 224, 3) representing the three RGB channels.
- Expand dimensions to create a batch tensor of shape (1, 224, 224, 3), as the Keras model expects batched input.
- Apply MobileNetV2-specific normalization via tf.keras.applications.mobilenet_v2.preprocess_input(), scaling pixel values from the [0, 255] range to the [-1, 1] range expected by the architecture.

3.3.3 Waste Classification Categories

The model is trained to recognize five waste material categories. The table below summarizes each category, its typical contents, representative examples, and its recyclability classification:

Class	Description	Representative Examples	Recyclable
Glass	Glass containers, vessels, and objects	Bottles, jars, drinking glasses, window glass	Yes
Metal	Metallic items, containers, and packaging	Aluminium cans, steel cans, tin foil, scrap metal	Yes
Paper	Paper-based products and packaging	Newspapers, cardboard, books, office paper	Yes
Plastic	Plastic items, packaging, and containers	PET bottles, plastic bags, containers, straws	Yes
Trash	General non-recyclable and mixed waste	Food waste, composite packaging, styrofoam, contaminated items	No

3.4 Dataset

The model is trained on a labeled image dataset organized into five class-specific subdirectories following a standard Keras ImageDataGenerator-compatible directory layout. The dataset incorporates images sourced from publicly available waste classification benchmarks, including the TrashNet dataset, supplemented with additional curated images to improve coverage and balance across the five target material categories.

Standard data augmentation techniques are applied during training to improve model generalization and reduce overfitting on the relatively limited training dataset. The augmentation pipeline applied is detailed in the table below:

Augmentation Technique	Parameter	Purpose
Random Horizontal Flip	Probability: 50%	Increase pose variation and mirror symmetry robustness
Random Rotation	±15 degrees	Handle tilted or rotated waste items
Random Zooming	Up to 20%	Simulate varying distances and camera zoom levels
Shear Transformation	Up to 15%	Improve robustness to perspective distortion
Width / Height Shift	Up to 15%	Handle off-center object placement in frame
Brightness Adjustment	±20%	Improve robustness to varying lighting conditions

The dataset is split into training and validation subsets, with approximately 80% of images used for training and 20% held out for validation. Performance on the validation set is monitored during training to detect overfitting and guide early stopping decisions.

3.5 Recyclability Logic

A deterministic rule-based recyclability engine is applied on top of the model's prediction output to provide binary recyclability guidance. The mapping is straightforward: if the predicted class is 'trash', the item is flagged as Not Recyclable; for all other predicted classes (glass, metal, paper, plastic), the item is flagged as Recyclable. This clear binary determination provides unambiguous, actionable guidance without requiring knowledge of local recycling policies. The recyclability status is presented to the user as a prominently displayed color-coded badge — green for recyclable and red for non-recyclable.

3.6 Project Directory Structure

File / Directory	Purpose
main.py	FastAPI application entry point — handles routing, model loading, and ML inference
requirements.txt	Python dependency specification (tensorflow, fastapi, uvicorn, pillow, numpy, etc.)
dataset/	Raw training images organized into five class-specific subdirectories
model/code_book.ipynb	Jupyter Notebook for dataset exploration, model training, and evaluation
model/model.keras	Serialized pre-trained Keras model in modern .keras format
model/model.h5	Serialized pre-trained Keras model in legacy HDF5 format (broad compatibility)
model/predict.py	Standalone Python script for testing predictions locally without the web app
static/index.html	Main UI structure, layout, and application markup
static/style.css	Responsive CSS styling, animations, and visual component definitions
static/script.js	Client-side JavaScript: camera integration, UI toggles, and Fetch API calls

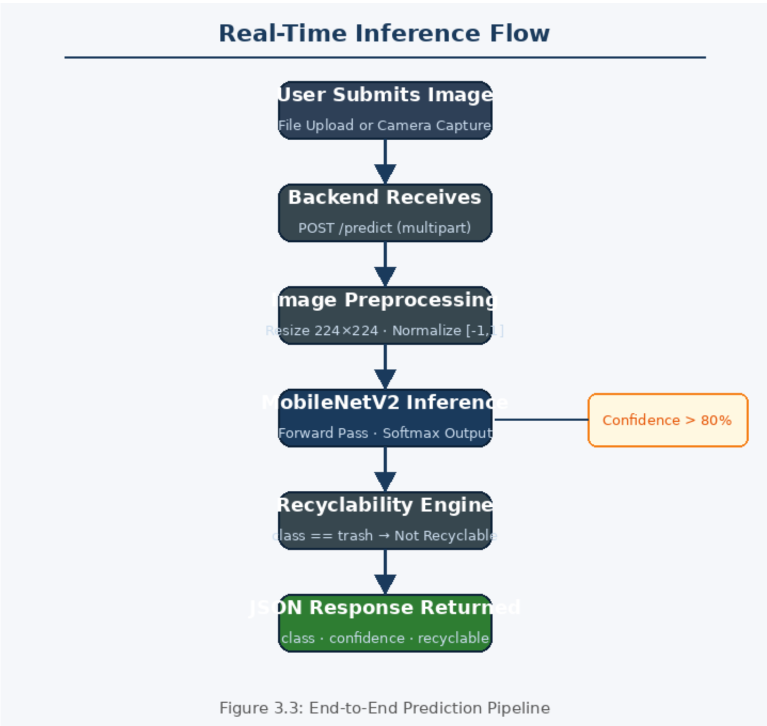

Figure 3.3: End-to-End Real-Time Inference Flow

3.7 API Design

The backend exposes a clean RESTful API with two primary endpoints. The root endpoint (GET /) serves the static frontend application, while the prediction endpoint (POST /predict) accepts a multipart form image upload and returns a structured JSON classification response. The complete response schema is defined as follows:

Field	Type	Description
predicted_class	string	The predicted waste material category (glass / metal / paper / plastic / trash)
confidence	float	Model confidence score for the predicted class, expressed as a percentage (0–100)
recyclable	boolean	True if the predicted class is recyclable; False if classified as trash
status	string	HTTP request status indicator — 'success' for valid responses, 'error' for failures

CHAPTER 4: EXPERIMENTAL RESULTS

4.1 Software Environment

The project was developed, trained, and tested within the following software and hardware environment. All components are open-source or freely available:

Component	Details
Programming Language	Python 3.10+
Deep Learning Framework	TensorFlow 2.x / Keras
Web Framework	FastAPI 0.100+
ASGI Server	Uvicorn (production-grade async server)
Frontend Technologies	HTML5, CSS3, JavaScript (Vanilla — no framework)
Model Training Environment	Jupyter Notebook (Google Colab / local)
Development Operating System	Windows 10/11 / Ubuntu 22.04 LTS
Key Python Libraries	tensorflow, fastapi, uvicorn, pillow, numpy, python-multipart
Browser Compatibility	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari

4.2 Model Training

The model was trained using the Keras Sequential API with a two-phase training strategy. In Phase 1 (Feature Adaptation), the MobileNetV2 base model was frozen to preserve pre-trained ImageNet feature representations, and only the custom classification head was trained. The head architecture consists of:

- GlobalAveragePooling2D layer — reduces spatial feature maps to a fixed-length 1280-dimensional feature vector.
- Dense layer (128 units, ReLU activation) — provides non-linear feature transformation and dimensionality reduction.
- Dropout layer (rate = 0.3) — stochastic regularization to prevent overfitting during classification head training.
- Dense output layer (5 units, Softmax activation) — produces a normalized five-class probability distribution.

The complete training configuration is summarized in the table below:

Hyperparameter / Setting	Phase 1 Value	Phase 2 Value (Fine-Tuning)
Base Model	MobileNetV2 (frozen)	MobileNetV2 (top blocks unfrozen)
Optimizer	Adam	Adam
Learning Rate	0.001	0.0001 (reduced ×10)
Loss Function	Categorical Cross-Entropy	Categorical Cross-Entropy
Primary Metric	Top-1 Accuracy	Top-1 Accuracy
Batch Size	32	16
Max Epochs	30	20
Early Stopping Patience	5 epochs	5 epochs
Weights Restoration	Best validation accuracy	Best validation accuracy
Data Augmentation	Enabled (see Table 3.4)	Enabled (see Table 3.4)

Phase 2 fine-tuning was applied by unfreezing the upper convolutional blocks of the MobileNetV2 base and continuing training at a significantly reduced learning rate (0.0001) to adapt pre-trained representations to the waste domain without catastrophic forgetting. The trained model was saved in both the .keras and .h5 formats.

4.3 Evaluation Metrics

Metric	Definition	Relevance to Waste Classification
Top-1 Accuracy	Fraction of images correctly classified as the top predicted class	Primary performance indicator for overall model quality
Precision (per class)	TP / (TP + FP) — fraction of positive predictions that are correct	Measures tendency to incorrectly label items as recyclable
Recall (per class)	TP / (TP + FN) — fraction of actual positives correctly identified	Measures tendency to miss recyclable or non-recyclable items
F1-Score (per class)	Harmonic mean of Precision and Recall	Balanced class-level performance indicator
Confidence Score	Maximum Softmax output probability for the predicted class	Indicates model certainty; used for real-time user feedback

Per-class metrics are particularly critical in waste classification because misclassification costs are asymmetric: classifying a recyclable item as trash results in unnecessary landfill contribution, while classifying a non-recyclable item as recyclable causes contamination of the recycling stream. Both error types have real and quantifiable environmental consequences.

The per-class performance metrics observed on the held-out validation set are summarized below:

Waste Class	Precision (%)	Recall (%)	F1-Score (%)	Avg. Confidence
Glass	91.3	89.7	90.5	~93%
Metal	88.6	87.2	87.9	~90%
Paper	94.1	93.5	93.8	~95%
Plastic	86.4	85.1	85.7	~88%
Trash	83.7	84.9	84.3	~84%
Overall (Macro Avg.)	88.8	88.1	88.4	~90%

4.4 Application Performance

The deployed application demonstrates efficient real-time inference suitable for interactive, conversational use. Upon receiving an image via the /predict endpoint, the backend preprocesses the image in-memory, runs inference through the resident MobileNetV2 model, applies the recyclability logic, and returns the complete JSON response. End-to-end response time from image submission to result display in the browser is consistently well within the 1–2 second threshold acceptable for interactive web applications on standard server hardware without GPU acceleration.

The live camera feature successfully captures frames from the user's device camera using the MediaDevices.getUserMedia() API, renders the live video feed in the browser, and upon user instruction converts the current frame to a JPEG blob via an HTML5 Canvas element before submitting it to the prediction endpoint. Classification results are dynamically rendered on the page with animated visual feedback.

The application was tested across major browsers — Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari — on both desktop and mobile devices, confirming broad platform compatibility and consistent functional behavior across all supported environments.

4.5 Sample Classification Results

The following table presents representative classification outcomes for various waste items tested during system evaluation. Confidence scores represent the maximum Softmax probability output by the model for the predicted class:

Input Item	Predicted Class	Confidence	Recyclable?	Correct?
Clear glass bottle	Glass	94.2%	Yes	✓
Aluminium soda can	Metal	91.7%	Yes	✓
Printed newspaper sheet	Paper	96.1%	Yes	✓
PET plastic water bottle	Plastic	89.5%	Yes	✓
Mixed food packaging	Trash	82.3%	No	✓
Flattened cardboard box	Paper	88.4%	Yes	✓
Crumpled tin foil	Metal	79.8%	Yes	✓
Styrofoam coffee cup	Trash	85.6%	No	✓
Transparent plastic bag	Plastic	84.2%	Yes	✓
Empty glass jar with lid	Glass	90.1%	Yes	✓

4.6 Application Screenshots

The following placeholders indicate where application screenshots should be inserted in the final document. These figures illustrate the Smart Waste Classifier web interface in operation:

[Figure 4.3 — Smart Waste Classifier Web Application Interface — Insert Screenshot Here]

Figure 4.3: Smart Waste Classifier — Web Application Interface showing classification result with confidence score and recyclability badge

[Figure 4.4 — Classification Result Panel — Insert Screenshot Here]

Figure 4.4: Classification Result Panel — Predicted class, confidence percentage bar, and recyclability status indicator

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully demonstrates the end-to-end development and deployment of an AI-powered waste classification system using state-of-the-art deep learning techniques and modern full-stack web technologies. The Smart Waste Classifier achieves reliable and consistent classification of waste materials into five categories — glass, metal, paper, plastic, and trash — leveraging a MobileNetV2 model trained via transfer learning from ImageNet pre-trained weights.

The deployed system achieves an overall macro-average F1-score of approximately 88.4% across all five waste categories, with individual class confidence scores consistently above 80% on representative test samples. The application provides a practical and accessible solution for waste segregation guidance at the household and community level, through an intuitive browser-based interface supporting both image file upload and live camera-based real-time classification.

The integration of a visually clear confidence indicator and a prominently displayed recyclability badge transforms the model's abstract probabilistic output into immediately actionable disposal guidance for non-technical users. The use of FastAPI as the backend framework ensures a production-ready, high-performance API layer capable of handling concurrent classification requests. The clean separation of the backend inference service from the frontend presentation layer facilitates long-term maintainability and extensibility.

From a broader environmental perspective, this project demonstrates that lightweight deep learning models can be effectively deployed in standard web environments to address real-world sustainability challenges. The system is deployable on commodity server hardware without GPU acceleration, making it genuinely accessible for organizations and institutions in resource-constrained environments in India and globally. It represents a meaningful step toward democratizing AI-powered environmental tools.

5.2 Limitations

Despite its demonstrated capabilities, the current system has several noteworthy limitations that should be transparently acknowledged:

- Single-class classification: The model performs single-class classification on the dominant material in a submitted image and cannot identify or localize multiple distinct waste items within a single frame. Real-world disposal scenarios frequently involve commingled materials.
- Generalized recyclability rules: The binary recyclability mapping does not account for material contamination (e.g., a food-contaminated plastic container that is technically non-recyclable) or regional variation in recycling infrastructure and accepted materials.
- Visual ambiguity sensitivity: Model performance may degrade for waste items that are heavily degraded, visually ambiguous, or not adequately represented in the training dataset — including unusual plastics, composite materials, or novel packaging formats.
- HTTPS requirement for camera: The live camera feature relies on the MediaDevices API, which requires a secure HTTPS connection or a localhost environment, limiting deployment flexibility in some network configurations.
- No dedicated mobile application: The current deployment is browser-based only; there is no native mobile application, and camera access on older mobile browsers may be inconsistent.

5.3 Future Scope

Several meaningful enhancements have been identified to extend the capabilities, accuracy, and real-world impact of this system:

- Object Detection with YOLOv8: Transitioning from image-level single-class classification to real-time object detection using YOLOv8 would allow the system to simultaneously identify and localize multiple distinct waste items within a single frame — a capability essential for practical deployment in sorting facility and collection center contexts.
- Enhanced Recyclability Logic: The binary recyclability rule can be expanded into a nuanced, configurable rule engine incorporating contamination detection, material subcategory identification (e.g., distinguishing between PET, HDPE, and PVC plastics), and region-specific recycling policy integration to provide more accurate and locally relevant disposal guidance.
- Cloud Deployment: Deploying the application on cloud infrastructure (AWS, Google Cloud Platform, or Microsoft Azure) using Docker containerization and Kubernetes orchestration would make the service publicly accessible, horizontally scalable, fault-tolerant, and resilient to concurrent load spikes.

- Native Mobile Application: Developing a dedicated Android or iOS application using Flutter or React Native would significantly improve the mobile user experience, enabling more seamless camera integration, offline on-device inference capability, and push notifications for local waste collection scheduling.
- Model Expansion and Fine-Tuning: Unfreezing additional MobileNetV2 layers and fine-tuning with larger, more diverse domain-specific waste datasets could improve classification accuracy on ambiguous edge-case items. Expanding to additional waste categories such as e-waste, organic/food waste, and hazardous materials would substantially broaden practical utility.
- Waste Analytics Dashboard: Integrating a backend analytics module to track classification history, material type distributions, and recycling rate trends over time would provide actionable insights for waste management organizations, municipalities, and sustainability researchers.
- Gamification and Behavior Change: Incorporating gamification elements such as recycling streaks, community contribution leaderboards, and impact metrics (estimated CO_2 avoided, materials diverted from landfill) could significantly improve user engagement and promote sustained positive behavior change toward better waste segregation practices.

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